

Intersection Theory

Labix

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Abstract

Contents

1 Intersection Theory

1.1 The Divisor of a Rational Function

Definition 1.1.1 (The Order of a Function at a Subvariety)

Let k be a field. Let X be a scheme of finite type over k . Let W be an irreducible subvariety of X . Let $f/g \in K(W)^*$ for $f, g \in \mathcal{O}_X(W)$ and $g \neq 0$. Let $V \subseteq W$ be an irreducible subvariety of codimension 1. Define the order of f at V to be

$$\text{ord}_V(f/g) = l_{\mathcal{O}_X(W)}\left(\frac{\mathcal{O}_X(W)}{(f)}\right) - l_{\mathcal{O}_X(W)}\left(\frac{\mathcal{O}_X(W)}{(g)}\right)$$

Definition 1.1.2 (The Divisor of a Rational Function)

Let k be a field. Let X be a scheme of finite type over k . Let W be an irreducible subvariety of X . Let $f \in K(W)^*$. Define the divisor of f to be

$$\text{div}(f) = \sum_{\substack{V \text{ is a codimension 1} \\ \text{irreducible subvariety of } W}} \text{ord}_V(f) \cdot V$$

1.2 The Chow Groups

Definition 1.2.1 (The Group of n -Cycles)

Let k be a field. Let X be a scheme of finite type over k . Let $n \in \mathbb{N}$. Define the group of n -cycles on X to be the free abelian group

$$Z_n(X) = \mathbb{Z}\langle V \mid \begin{smallmatrix} V \text{ is an } n\text{-dimensional} \\ \text{irreducible subvariety of } X \end{smallmatrix} \rangle$$

Lemma 1.2.2

Let k be a field. Let X be a scheme of finite type over k . Let W be an irreducible subvariety of X . Let $f \in K(W)^*$. Then $\text{div}(f) \in Z_{\dim(W)-1}(X)$.

Definition 1.2.3 (Cycles Rationally Equivalent to Zero)

Let k be a field. Let X be a scheme of finite type over k . Let $V \in Z_n(X)$. We say that V is rationally equivalent to 0 if there exists finitely many $n+1$ -dimensional irreducible subvarieties $W_1, \dots, W_r$ of X and $f_i \in K(W_i)^*$ such that

$$V = \sum_{i=1}^n \text{div}(f_i)$$

Proposition 1.2.4

Let k be a field. Let X be a scheme of finite type over k . Let $V \in Z_n(X)$ be a cycle. Then V is rationally equivalent to 0 if and only if there exists $(n+1)$ -dimensional irreducible subvarieties $W_1, \dots, W_r$ of $X \times \mathbb{P}_k^1$ such that the projection maps $p_i : W_i \rightarrow \mathbb{P}_k^1$ are dominant and $V = \sum_{i=1}^r [p_i^{-1}(0)] - [p_i^{-1}(\infty)]$

Definition 1.2.5 (The Chow Group)

Let k be a field. Let X be a scheme of finite type over k . Define the n th Chow group of X to be

$$\text{CH}_n(X) = \frac{Z_n(X)}{\sim}$$

where we say that $V_1 \sim V_2$ if $V_1 - V_2$ is rationally equivalent to 0. In this case, we say that V_1 and V_2 are rationally equivalent.

1.3 Pushforward and Pullback of Cycles

Definition 1.3.1 (The Push Forward Map)

Let k be a field. Let X, Y be schemes of finite type over k . Define the push forward map $f_* : Z_n(X) \rightarrow Z_n(Y)$ by

$$f_*(V) = \deg(V/f(V))f(V)$$

where

$$\deg(V/f(V)) = \begin{cases} [k(V) : k(f(V))] & \text{if } \dim(V) = \dim(f(V)) \\ 0 & \text{otherwise} \end{cases}$$

Proposition 1.3.2

Let k be a field. Let X, Y be schemes of finite type over k . Then the push forward map descends to a well defined homomorphism $f_* : \text{CH}_n(X) \rightarrow \text{CH}_n(Y)$.

Proposition 1.3.3

Let k be a field. Let X be a scheme of finite type over k . Let $i : Y \hookrightarrow X$ be a closed subscheme of X . Let $j : X \setminus Y \hookrightarrow X$ be the inclusion map. Then there is an exact sequence

$$\text{CH}_n(Y) \xrightarrow{i_*} \text{CH}_n(X) \xrightarrow{j_*} \text{CH}_n(X \setminus Y) \longrightarrow 0$$